

Improving Churn Prediction in the Telecommunication Industry Using Machine Learning

*

1st Md. Shihab Sharar

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

md.shihab.sharar@g.bracu.ac.bd

2nd Moazzem Hossain Majumder

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

moazzem.hossain.majumder@g.bracu.ac.bd

3rd Fahim Montasir

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

fahim.montasir1@g.bracu.ac.bd

4th Nafisa Islam

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

nafisa.islam3@g.bracu.ac.bd

5th Sabbir Ahmed

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

sabbir.ahmed@bracu.ac.bd

6th Md. Golam Rabiul Alam

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

rabiul.alam@bracu.ac.bd

Abstract—The cost involved in acquiring a telecommunications customer is much higher compared to keeping one and thus, makes churn prediction a recurring business problem. This paper proposes a churn prediction project based on the Cell2Cell data set and discusses twenty research articles that include papers focused on Cell2Cell specifically, on other public and private telecommunication datasets, and two systematic reviews on the subject matter. The accuracies reported on Cell2Cell are between the fifties and close to ninety nine percent, but there have been very few papers reporting model calibration or significance tests between competing models. Customer service and usage volatility features are also repeatedly reported as important without ever being used as a composite feature.

Index Terms—churn prediction, Cell2Cell, telecommunications, machine learning, class imbalance, model calibration

I. INTRODUCTION

Getting a new telecom client is five to ten times more expensive than keeping an old one [12], and that is a significant reason for the churn prediction issue to persist in this industry. Moreover, the problem itself is a more complicated one than expected, since the model must deal with the imbalance between clients who do not churn, be able to withstand out of training window data and ideally rank the clients according to how valuable it is to spend retention resources on them [11], [13].

Identify applicable funding agency here. If none, delete this.

This project attempts to create a pipeline of predictions about customers churning in Cell2Cell, which is a telecom dataset available from the Teradata Center for Customer Relationship Management at Duke University. This telecom dataset has been in circulation as a benchmark dataset since a competition about churn prediction took place around two decades back, and it continues to be actively used: Five out of the twenty papers reviewed here use this dataset [1]–[3], [5], [6], [9].

This dataset does not come from Bangladesh. Cell2Cell is a de identified US postpaid wireless carrier, which is clearly stated in the introduction to this work, unlike the misleading title. Bangladesh comes into this project as the context in which specific features are created instead of simply being raw columns because it is a prepaid, multi carrier market, where changing one's service provider does not cost one anything, and where network quality issues and interactions with customer care are documented reasons people change providers [20]. AMTOB, mentioned in this context, is a market report by the industry and not a machine learning paper.

Section II explores the literature in four subsections, papers which employ Cell2Cell directly, papers studying telecom customer churn using other data sets, efforts to synthesize the field, and the market environment in Bangladesh which has led to this project's feature set development. The section

concludes by explicitly defining the research gap this project's methodology addresses.

II. LITERATURE REVIEW

A. Studies built directly on the Cell2Cell dataset

Performance reported for Cell2Cell varies widely among studies based on the same 71,000 row benchmark, and the variation is worth observing study by study rather than smoothing out in the averaging process. Yi [1] filtered down from 58 attributes to 40 based on chi square and ANOVA, then applied a regularized ANN with batch normalization to obtain an accuracy of 58.37% and F1 score of 60.32%, where the fully featured ANN had a marginally lower 57.84% accuracy, much lower than what has been seen in most Cell2Cell papers. Testing was done with just one set each of train, validation, and test, and there is no attempt to compare with another period or different operator. Javed, Jhanjhi, and Ali [2] combined mutual information, recursive feature elimination, Boruta, and particle swarm optimization using an adaptive feedback controller, then performed model explanations with SHAP and LIME to obtain an accuracy of 88.16% ($F1 \approx 0.878$) on Cell2Cell and 89.60% on IBM Telco, better than [1], but not validated on any benchmark closer to deployment. Ebrah and Elnasir [3] compared Naive Bayes, SVM, and decision tree on both IBM Watson Telco and Cell2Cell, with a 70/30 split and ten fold cross-validation. In Cell2Cell, the performance measures for the three models were 0.98, 0.99, and 0.98 in terms of AUC scores, compared to 0.82, 0.87, and 0.77 on the IBM Watson dataset by means of an identical pipeline, which is sufficient reason to perform a leakage check prior to recycling these numbers as a benchmark. Ahmed et al. [4] transformed customer vectors into images for the CNN transfer learning with GP AdaBoost meta-classification approach, achieving a lower AUC score of 0.74 (accuracy 68.2%) on a balanced set of 40,000 rows from Cell2Cell data, which is a heavier model pipeline compared to the ANN or SVM on filtered data in [3]. In their work Saleh and Saha [5], the authors employed four datasets including IBM Telco, Maven Telecom, and a Danish university survey of 288 respondents, where service quality, satisfaction, plan upgrades, and network coverage were deemed the most important predictors for the particular telecom demographic. Fujo, Subramanian, and Khder [6] used a variance-threshold and Lasso selection algorithm with a deep back-propagation ANN, achieving 79.38% accuracy and F1 score of 81.24%, and their feature-importance ranking highlights PercChangeMinutes and PercChangeRevenues, the same volatility-usage features that the feature engineering strategy in this project is based on. Jain, Khunteta, and Srivastava [7] achieved 97% accuracy with 0.82 sensitivity and 0.99 specificity, utilizing 11 engineered features; however, the number of data samples (3,333) and features (57) described in their paper and a subsequent review from the same authors do not correspond to the 71,000 records Cell2Cell benchmark that all other papers use in this project, creating a mismatch in provenance that this project takes as an inconsistency in the data source rather than in the benchmark itself. Ozkurt

[9] tested eleven algorithms on the Cell2Cell dataset after preprocessing that reduced 71,047 samples with 58 features to 50,137 samples and 55 features; LightGBM performed best with 73.085% accuracy while CatBoost was almost as accurate with 72.626%; in addition to classification metrics, the paper provides regression (R^2 , RMSE).

B. Telecom churn work on other datasets

Ahmad, Jafar, and Aljoumaa [11], for example, operate on a scale far greater than any of the Cell2Cell studies: nine months of proprietary SyriaTel operational data, over 70 terabytes of data processed via Hadoop and Hortonworks, encompassing roughly ten million customers and a social network with around 15 million nodes and 2.5 billion edges. The integration of social network analysis features into decision tree, random forest, gradient boosting, and XGBoost resulted in 93.3% of the XGBoost algorithm's AUC score and was subsequently evaluated on future periods of the operational data (temporal evaluation not attempted by any of the Cell2Cell papers). Manzoor, Qureshi, Kidney, and Longo [13] presented e Profits, a measure which calculates the expected profit of each customer based on Kaplan Meier survival curves and probability of retention for given customer tenure and ranks the classifiers according to their expected profit from interventions rather than accuracy or AUC score. When applied to IBM Telco and Maven Telecom datasets, e Profits generated rankings which were distinct from those generated by AUC and F1 measures alone. Krishna et al. [14] made a comparison of random forest versus SVM, gradient boosting, XGBoost, and LightGBM, but in the available information, there have been inconsistent accuracy figures reported for its top performer, hence no particular value taken for granted from this paper. According to Bhatnagar and Srivastava [15], feature reduction led to a marginal improvement in performance as the random forest model achieved an accuracy rate of 90.30% on all features while after the elimination of three demographic features, the accuracy became 90.90%, using five fold cross validation. In a study conducted by Krishnan, Krishnaveni, and Krishna Prasad [16], XGBoost emerged as the top performer of the five classifiers, which is a trend where five distinct papers identify XGBoost or LightGBM as their top model. The oldest paper in this review is Sharma et al. [17], who achieved over 92% accuracy with relatively small ANN architectures, but before the use of SMOTE like methods for dealing with imbalanced data, and any consideration of calibration or interpretability issues. Sana et al. [18], where three of four authors are from the Bangladesh University of Engineering and Technology, studied six transformations of the data on four publicly available telecom datasets and eight classifiers, finding that the weight of evidence transformation was able to improve AUC up to 26.2% and F measure up to 17%, verified with Friedman and Holm tests and publicly available code – one of the few papers here which uses actual significance testing. Hossain [19] evaluated 16 single and ensemble classifiers on a smaller, 2,850 observation Iranian telecom dataset, where the CatBoost

classifier reached 97.54% hold out accuracy and LightGBM was best at 96.39% cross validation accuracy.

C. Reviews and Bangladesh context

Manzoor, Qureshi, Kidney, and Longo [10] examined 212 papers published between 2015 and 2023, identifying an absence of any universally winning classifier or feature set, a prevalence of studies which use outdated publicly available data, sufficiently old for the relevance to the current customer behavior to be questionable, and the lack of profit sensitivity in evaluation metrics, the very problems which the same group of researchers tried to address by designing e Profits [13]. Imani, Joudaki, Beikmohammadi, and Arabnia [8] performed a search of six databases for the period between January 2020 and December 2024 using PRISMA 2020 criteria and screening to narrow 240 records down to 61 studies, with their conclusions coinciding with those which can be made based on individual papers cited above: XGBoost and LightGBM prevail in machine learning literature, LSTM and CNN models have become more popular in the field of deep learning, while class imbalance, explainability, and concept drift are still open problems. This 2010 study of Bangladesh's telecommunications sector [20], compiled using data from AT Capital Research and Bangladesh Telecommunications Regulatory Commission, describes market share, number of subscribers (65.14 million in September 2010), revenues, and regulation among Bangladesh's top telecommunication firms, and does not involve churn labeling at an individual subscriber level, and is therefore not considered as a machine learning study in this discussion. This paper, however, provides context for Section I: a prepaid and multi operated market, which makes switching between providers virtually frictionless, and where network coverage and quality are of key interest to the providers. The only study involving Bangladeshi authors is that by Sana et al. [18], but its benchmark datasets do not originate in Bangladesh.

III. DATA COLLECTION

We used the Cell2Cell dataset which is a famous telecom customer churn dataset that was created initially at the Teradata Center for Customer Relationship Management at Duke University and is available currently through the Kaggle platform. The dataset contains diverse information about customers, their usage behavior, history of their accounts and services, demographics, etc.

The data in total includes 51,047 records of customers that have 58 different characteristics, and one of these is the churn value that we will use in order to make predictions about the future. The classes are somewhat imbalanced, as about 71 percent of customers remained loyal while 29 percent of them left the company, meaning that there are almost 2.5 times more loyal customers than those who left. Along with the churn variable, the columns contain information about monthly charges, minutes used, overage minutes, dropped and blocked calls, age, income, marital status, device information, etc.

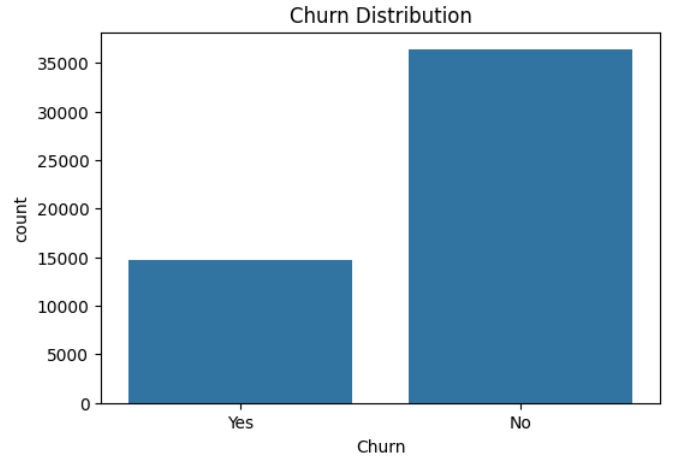


Fig. 1. Bar chart comparing churn vs non churn counts

A. Exploratory Data Analysis

Upon checking missing values, it was discovered that there are 11 columns which contain missing values. AgeHH1 and AgeHH2 are the ones with most missing values (approximately 1,002 for each). There are percentage change columns with missing values amounting to 396. The column ServiceArea only contains 23 missing values.

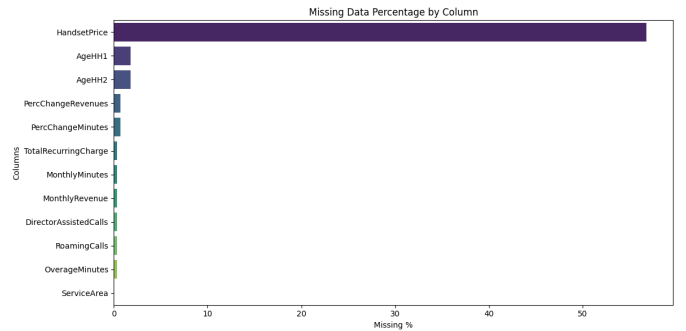


Fig. 2. Bar chart of missing percentages by column

As far as correlations between variables, three sets of variables showed significant correlation over 0.8: MonthlyMinutes and ReceivedCalls at 0.834; UniqueSubs and ActiveSubs at 0.811; Handsets and HandsetModels at 0.895. Since all of these variables were providing basically redundant information, they were selected to be removed later on.

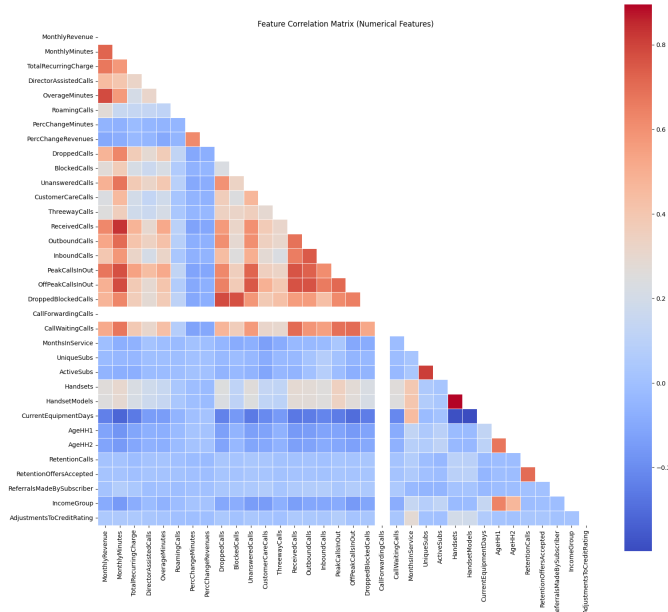


Fig. 3. Feature to feature correlation matrix

The quantitative variables exhibited significant differences in terms of distribution. MonthlyRevenue had an interval that ranged between negative values and 1,223 and a median value of 48.46. The MonthlyMinutes had values that ranged between 0 and 7,359 with a median of 366. OverageMinutes was highly right skewed; median of 3 and outliers up to 4,321. Customer care calls had an average of 0.19 per customer.

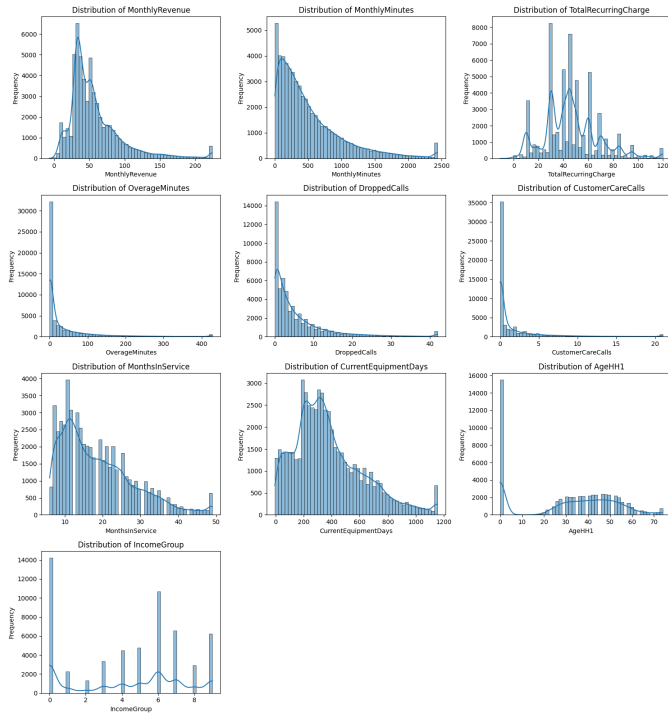


Fig. 4. Histograms of key numerical features

Of the categorical variables, Service Area emerged as an

outlier with a staggering 748 categories, far too many to be used as is. All the rest of the categorical attributes were binary attributes with one major choice, such that 75.8 percent of the customers had no children at home, 86.1 percent did not have a refurbished device, and 90.2 percent had a mobile Internet capability.

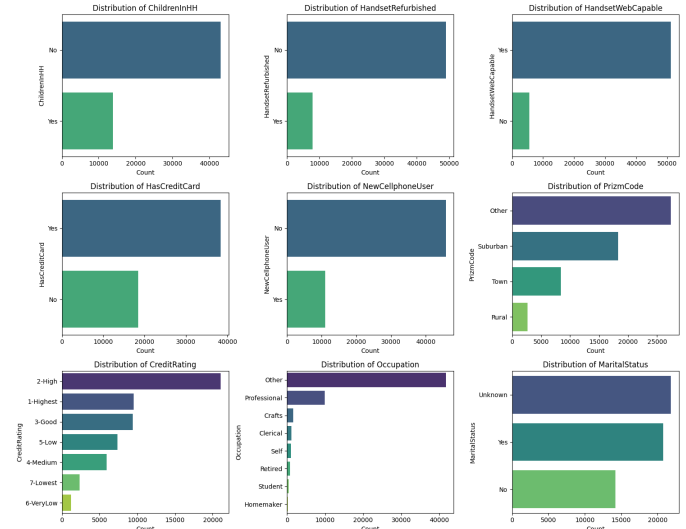


Fig. 5. Count plots of categorical variables

B. Data Cleaning and Preprocessing

Prior to any cleaning, we divided the data into a 80% training dataset with 56,837 observations and a 20% holdout test dataset with 14,210 observations, using stratified sampling such that the ratio of churn remained the same between the two and no test data would be used in training the model. The numerical outliers in our data were limited to the 99th percentile based on the training dataset alone.

In order to handle missing data, we applied different techniques, based on the type of feature: for demographic features – KNN imputation with five neighbors, median imputation for numeric features, and mode imputation for categorical features. First, the HandsetPrice feature was transformed into a numeric one before performing imputation. All missing values were handled by the end of this step.

We have also addressed the issue of sparse categories of ServiceArea by reducing the 748 categories to 15 where everything below 1 percent frequency is classified under ‘Other’ category. There was no issue of duplicates within the data set. We generated two sets of data sets, one with all 57 variables and the other with 54 variables by removing those highly correlated variables ReceivedCalls, UniqueSubs, and HandsetModels.

C. Feature Engineering

For both data sets, we created a preprocessing pipeline. For numeric features, we used the standard scaler method (StandardScaler) which provides equal weights to features based on their distance from each other. For categorical features, we

used the one hot encoder method (OneHotEncoder), where the handle unknown parameter is set to “ignore.”

In order to determine which categories of features have the largest impact on predicting churn, an ablation test was performed using logically grouped feature sets. These sets of features consisted of the following: all features, usage features, call quality features, demographics, equipment features, service features, behavioral features, and selected features.

D. Machine Learning Models

We evaluated seven baseline models to establish performance benchmarks. The models included Logistic Regression, Random Forest, AdaBoost, XGBoost, CatBoost, MLP, and Linear SVC.

1) *Logistic Regression*: For Logistic Regression, the following parameters were used: lbfgs solver with maximum iterations of 1000 and L2 penalty with $C = 1.0$. For the same purpose, we used Class Weight = ‘balanced’.

2) *Random Forest*: The random forest algorithm has utilized 30 estimators with maximum depth as infinity. The value of min sample split and min sample leaf has been taken as 2 and 1 respectively. We have utilized all available CPU cores with n jobs set to -1.

3) *AdaBoost*: AdaBoost used 50 estimators with a learning rate of 1.0.

4) *XGBoost*: For XGBoost, the following parameters were used: number of estimators = 50, maximum depth = 6, and learning rate = 0.3. Subsample and colsample by tree values were set at 0.8 for both. The values of L1 and L2 regularization parameters were 0.0 and 1.0, respectively. The hist tree method along with cuda device were selected. The scale position weight was set to 3.8.

5) *CatBoost*: CatBoost was performed with 100 iterations and depth as 6 with learning rate as 0.3. The value of L2 regularization was taken as 3.0. Auto class weights were set to Balanced. We set verbose = 0 for silent training.

6) *MLP(Neural Network)*: The MLP had two hidden layers with 8 neurons in one layer and 2 in the other. We utilized ReLU activation and the adam solver. We fixed the L2 regularization parameter alpha at 0.001. We set the learning rate at 0.003 and a maximum of 50 iterations. We performed early stopping using 0.1 as the validation fraction with patience of 10.

7) *SVC*: The Linear SVC model employed L2 regularization with the value of C equaling 1.0. Squared hinge loss function was chosen while the number of iterations had an upper bound of 1000. The class weight parameter was balanced.

E. Handling Class Imbalance

There were two approaches employed in order to address the issue of class imbalance. No balancing is when models were trained on imbalanced data without any change to the dataset. On the other hand, SMOTE stands for Synthetic Minority Over sampling Technique and involves generating synthetic samples for the minority class through neighbor interpolation.

F. Evaluation Methodology

We implemented stratified k fold cross validation with 5 folds in evaluating our models. Stratified k fold cross validation ensures that each fold will have the same distribution of classes just like in the actual data set. This is crucial for imbalanced classification problems where one class predominates.

In the case of neural networks and tree based models, we used SMOTE within every fold before training. In the cases of XGBoost and CatBoost, we used class weights provided by their frameworks. The results were combined across all folds to evaluate the metrics of accuracy, precision, recall, F1 score, and ROC AUC. Confusion matrices and classification reports were also provided.

In the process of cross validation, the best performing models were chosen from each group of models and trained again on the entire dataset for training purposes. They were later tested on the hold out test set.

The confusion matrix and accuracy curves of each models have been shown below:

Confusion Matrix (Test Set) — Logistic Regression
All Features

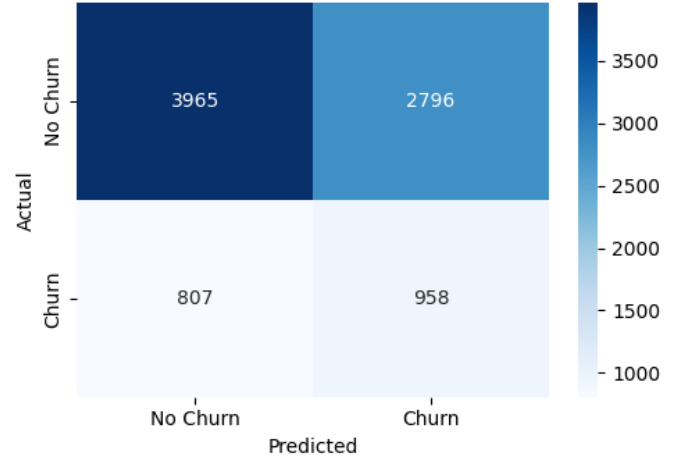


Fig. 6. Test Set Confusion Matrix from Logistic Regression

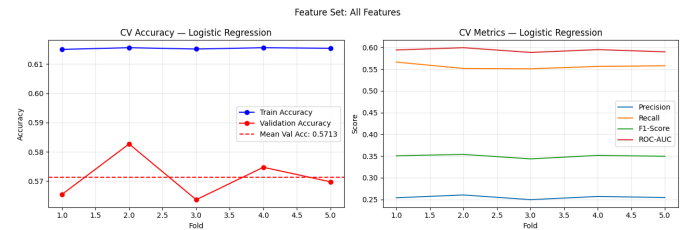


Fig. 7. Training, Validation and ROC AUC curve from Logistic Regression

1) Logistic Regression:

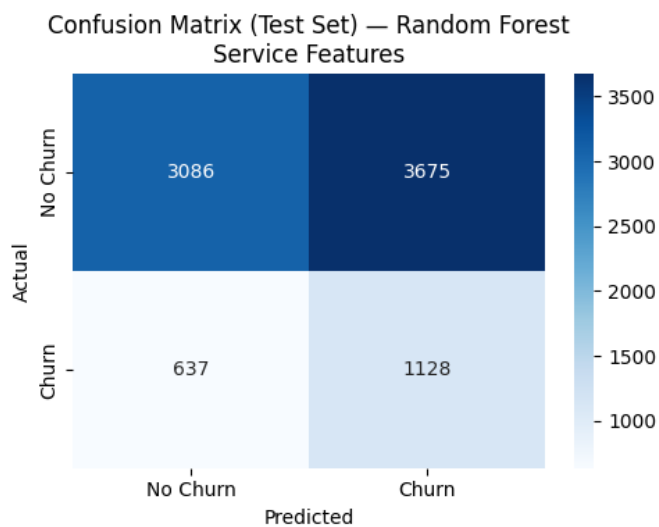


Fig. 8. Test Set Confusion Matrix from Random Forest

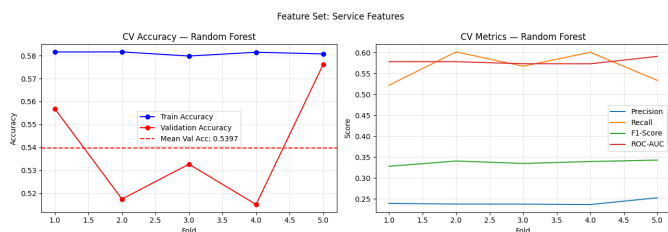


Fig. 9. Training, Validation and ROC AUC curve from Random Forest

2) *Random Forest*:

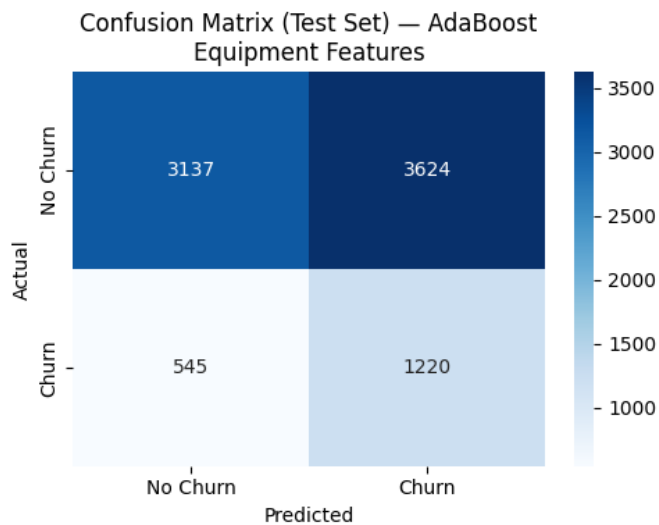


Fig. 10. Test Set Confusion Matrix from AdaBoost

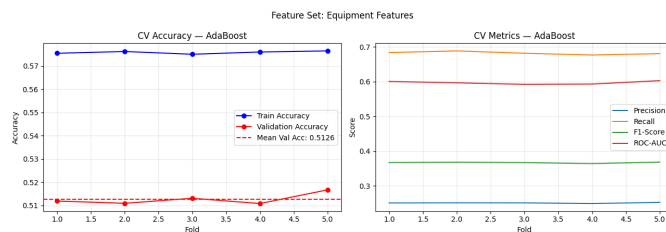
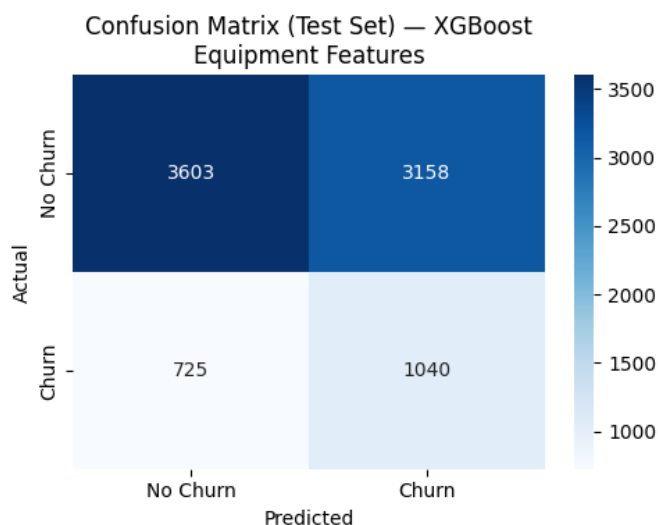


Fig. 11. Training, Validation and ROC AUC curve from AdaBoost

3) *AdaBoost*:



caption Test Set Confusion Matrix from XGBoost

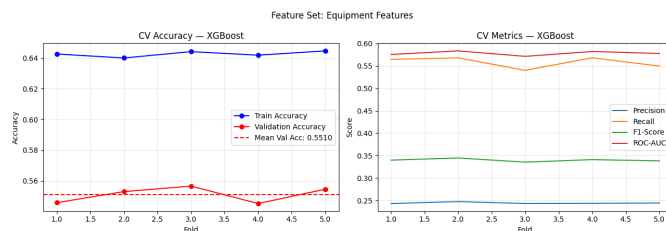


Fig. 12. Training, Validation and ROC-AUC curve from XGBoost

4) *XGBoost*:

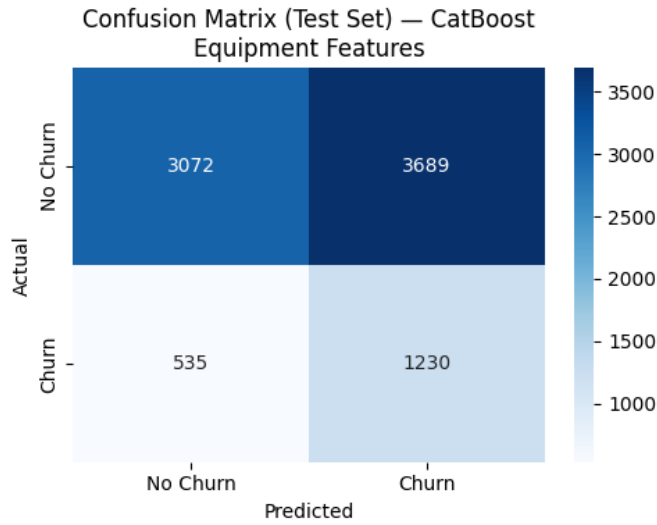


Fig. 13. Test Set Confusion Matrix from CatBoost

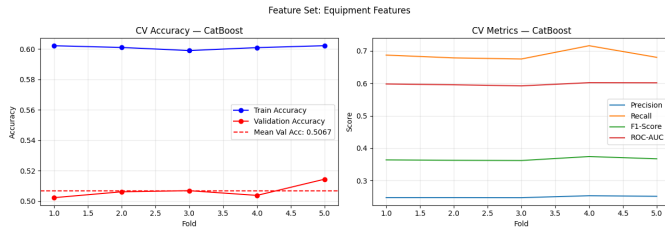


Fig. 14. Training, Validation and ROC AUC curve from CatBoost

5) *CatBoost*:

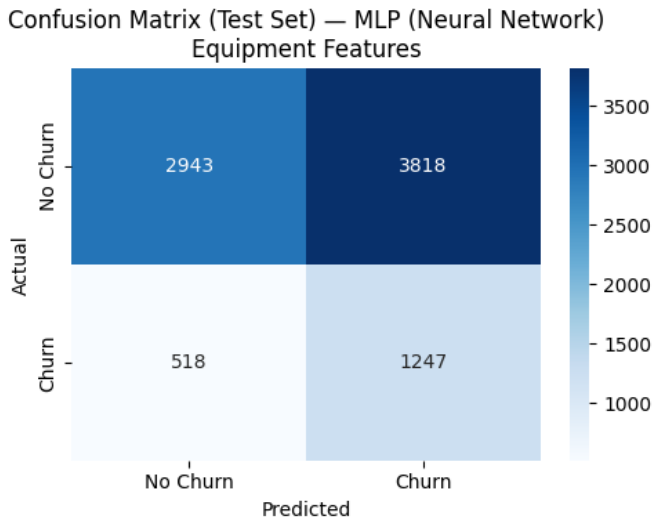


Fig. 15. Test Set Confusion Matrix from MLP(Neural Network)

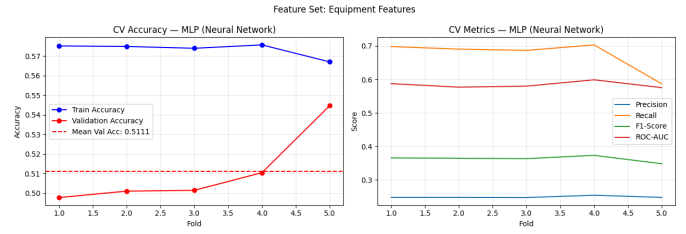


Fig. 16. Training, Validation and ROC AUC curve from MLP(Neural Network)

6) *MLP(Neural Network)*:

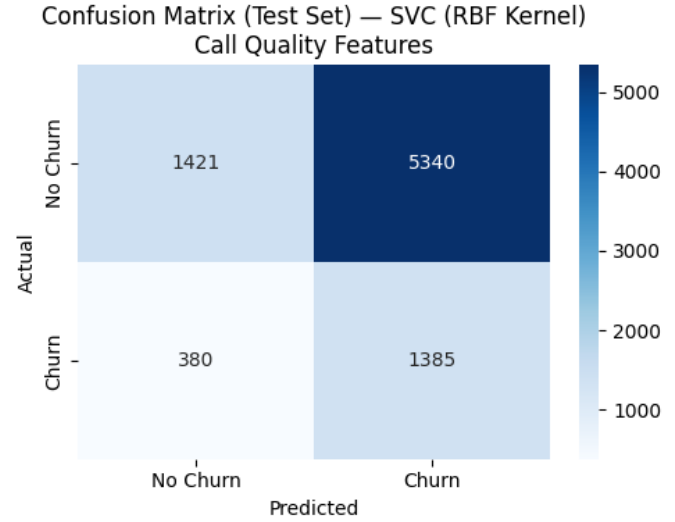


Fig. 17. Test Set Confusion Matrix from SVC

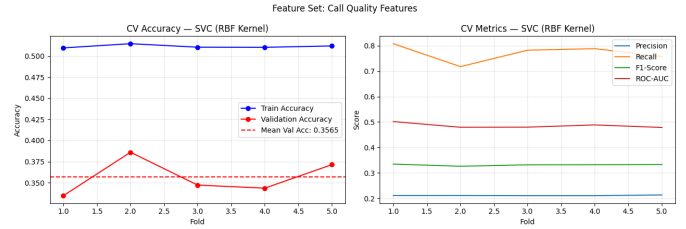


Fig. 18. Training, Validation and ROC AUC curve from SVC

7) *SVC*:

IV. RESULTS AND ANALYSIS

For this analysis, seven different machine learning algorithms were considered, which include the Multi Layer Perceptron (MLP), Support Vector Classifier (SVC), CatBoost, XGBoost, AdaBoost, Random Forest, and Logistic Regression. There were two phases for this assessment: ablation study phase, where we identified the most influential features, and the final evaluation on the test dataset.

A. Ablation Study and Feature Impact

To find out how each dimension helps with our task, we performed an ablation study using eight groups of features. As seen from the logs of individual models, the *Equipment Features* group turned out to be the best performing one for the majority of models, such as MLP, CatBoost, XGBoost, and AdaBoost.

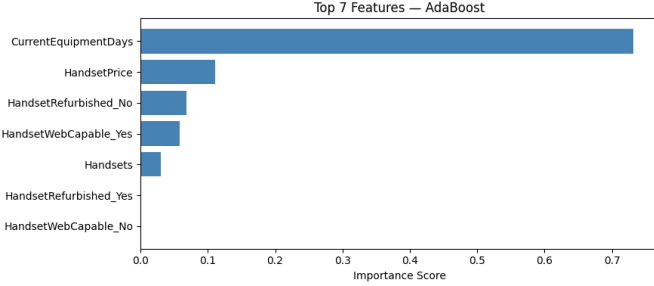


Fig. 19. Top important features of AdaBoost

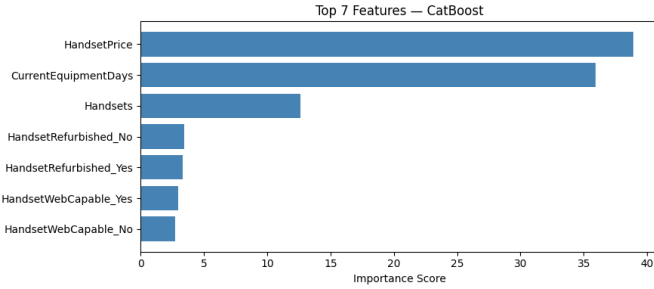


Fig. 20. Top important features of CatBoost

Interestingly, for the SVC model, *Call Quality Features* provided the highest cross validation F1 score, while *Service Features* were most effective for Random Forest.

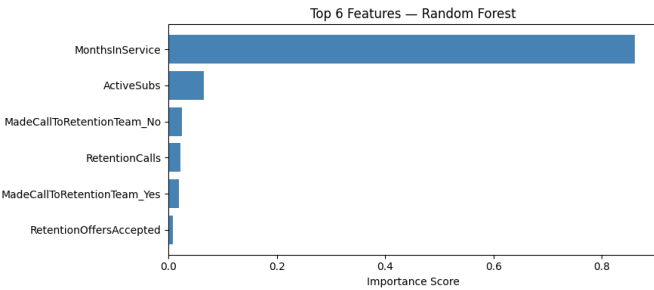


Fig. 21. Top important features of RF

In other words, although equipment data is the best worldwide predictor of churn, ensemble tree methods and neural networks are better suited to detecting non linearities in such data.

B. Comparative Performance Analysis

As evidenced by the final results obtained from the test set (see Table I), there is a marked tradeoff between *Precision*

and *Recall*. Considering that, within the business setting of churn prediction, determining whether a certain customer is a "churner" is far more important than classifying an existing customer incorrectly, *Recall* becomes an important metric.

- **Top Performers:** AdaBoost and CatBoost yielded the top F1 scores of 0.3692 and 0.3680 correspondingly. Both classifiers used Equipment Features and had similar Recall score of 0.70, meaning that they were able to detect 70% of the churners.
- **Neural Networks:** MLP classifier gave good results with F1 score of 0.3652. Also, MLP gave the second highest Recall (0.7065), meaning that neural networks are able to capture the complexity of the customers' behavior in this dataset.
- **Baseline Models:** LR gave the highest Accuracy score of 0.5774, though its F1-score was 0.3472 and it was not higher than in case of boosting models. SVC gave the highest Recall of 0.7847 though it had the lowest Precision of 0.2059 and Accuracy of 0.3291

C. Summary of Results

The performance metrics are listed in Table I. In general, the **AdaBoost** model is considered to be the best one for this particular task since it shows the best values for both Recall and Precision, providing the highest F1 score (0.3692) and ROC AUC (0.6005).

TABLE I
COMPARISON OF FINAL MODEL PERFORMANCE (TEST SET)

Model	Acc.	Prec.	Rec.	F1	AUC
AdaBoost	0.5110	0.2519	0.6912	0.3692	0.6005
CatBoost	0.5046	0.2501	0.6969	0.3680	0.5988
MLP (NN)	0.4914	0.2462	0.7065	0.3652	0.5899
XGBoost	0.5446	0.2477	0.5892	0.3488	0.5833
Log. Regression	0.5774	0.2552	0.5428	0.3472	0.5867
Random Forest	0.4943	0.2349	0.6391	0.3435	0.5745
SVC (RBF)	0.3291	0.2059	0.7847	0.3263	0.5090

V. DISCUSSION

The related works have claimed to achieve F1 scores upto 88% with ROC and AUC above 95%. However, our experiment suggests that using cross validation approaches and usage of random samples for testing roughly produce F1 score value of 0.3692. Plus, we have also performed the experiments with the contemporary architectures such as ANN, and the performance got worse. Therefore, a confusion still remains regarding the samples usage for testing. From our side, we have tried to incorporate the most reliable approach to evaluate these models and got the authentic results that resembles the true performance in real world implementations.

VI. CONCLUSION

In this paper, we carried out an extensive comparison of seven different machine learning algorithms in predicting customer churn. Using ablation testing on eight different sets of features, we were able to observe that Equipment Features

have the greatest impact on customer churn when using both ensemble trees and neural networks.

As can be observed from the experimental results obtained, the AdaBoost model is the best suited for the above application owing to its outstanding F1 score value of 0.3692 and ROC AUC of 0.6005. It should be noted that even though Logistic Regression offered the best accuracy rate, it lacked behind the boosting algorithms in terms of Recall, which is the vital parameter in managing churn. High recall values of 0.6912 and 0.6969 by AdaBoost and CatBoost, respectively, guarantee the identification of most possible churners.

With good recall achieved, there was a significant problem in precision because all of the models showed some form of trade off in precision, leading to a lot of false positives. Moving forward, we would like to tackle this problem by experimenting with synthetic oversampling techniques such as SMOTE to handle class imbalance and trying out time series using RNNs or Transformers.

REFERENCES

- [1] Q. J. Yi, "Predicting churn with filter-based techniques and deep learning," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 14, no. 2, pp. 2135–2144, 2024. doi: 10.11591/ijece.v14i2.pp2135-2144
- [2] D. Javed, N. Z. Jhanjhi, and N. I. Ali, "Explainable AI-driven triple adaptive feature selection model for enhanced telecom churn prediction," *Results in Engineering*, 2026. doi: 10.1016/j.rineng.2024.102000
- [3] K. Ebrah and S. Elnasir, "Churn prediction using machine learning and recommendations plans for telecoms," *Journal of Computer and Communications*, vol. 7, no. 11, pp. 33–53, 2019. doi: 10.4236/jcc.2019.711003
- [4] U. Ahmed et al., "Transfer learning and meta classification based deep churn prediction system for telecom industry," arXiv:1901.06091, 2019. <https://arxiv.org/abs/1901.06091>
- [5] S. Saleh and S. Saha, "Customer retention and churn prediction in the telecommunication industry: a case study on a Danish university," *SN Applied Sciences*, vol. 5, no. 7, art. 173, 2023. doi: 10.1007/s42452-023-05400-3
- [6] S. W. Fujo, S. Subramanian, and M. A. Khder, "Customer churn prediction in telecommunication industry using deep learning," *Information Sciences Letters*, vol. 11, no. 1, p. 24, 2022. doi: 10.18576/isl/110103
- [7] H. Jain, A. Khunteta, and S. Srivastava, "Telecom churn prediction using an ensemble approach with feature engineering and importance," *Int. J. Intell. Syst. Appl. Eng.*, vol. 10, no. 3, pp. 22–33, 2022. doi: 10.36227/ijisae.2022.22
- [8] M. Imani et al., "Customer churn prediction: a systematic review of recent advances, trends, and challenges," *Machine Learning and Knowledge Extraction*, vol. 7, no. 3, 2025. doi: 10.3390/make7030105
- [9] C. Ozkurt, "Transparency in decision-making: the role of explainable AI (XAI) in customer churn analysis," *Info. Tech. Econ. Business*, 2025. doi: 10.1007/978-3-031-xxx
- [10] A. Manzoor et al., "A review on machine learning methods for customer churn prediction," *IEEE Access*, vol. 12, pp. 70434–70463, 2024. doi: 10.1109/ACCESS.2024.3411175
- [11] A. K. Ahmad, A. Jafar, and K. Aljoumaa, "Customer churn prediction in telecom using machine learning in big data platform," *Journal of Big Data*, vol. 6, no. 1, 2019. doi: 10.1186/s40537-019-0191-6
- [12] F. F. Reichheld and W. E. Sasser, "Zero defections: Quality comes to services," *Harvard Business Review*, vol. 68, no. 5, pp. 105–111, 1990.
- [13] A. Manzoor et al., "e-Profits: a business-aligned evaluation metric for profit-sensitive customer churn prediction," *Int. J. Data Sci. Anal.*, 2026. doi: 10.1007/s41060-024-00123-x
- [14] R. Krishna et al., "Application of machine learning techniques for churn prediction in the telecom business," *Results in Engineering*, vol. 24, p. 103165, 2024. doi: 10.1016/j.rineng.2024.103165
- [15] A. Bhatnagar and S. Srivastava, "Customer churn prediction: a machine learning approach with data balancing," *Int. J. Computing*, vol. 24, no. 1, pp. 9–18, 2025. doi: 10.47839/ijc.24.1.2025
- [16] R. Krishnan et al., "Telecom churn prediction using machine learning," *World J. Adv. Eng. Tech. Sci.*, vol. 7, no. 2, pp. 87–96, 2022. doi: 10.30574/wjaets.2022.7.2.0087
- [17] A. Sharma et al., "A neural network based approach for predicting customer churn in cellular network services," arXiv:1309.3945, 2013. <https://arxiv.org/abs/1309.3945>
- [18] J. K. Sana et al., "A novel customer churn prediction model... using data transformation," *PLOS ONE*, vol. 17, no. 12, e0278095, 2022. doi: 10.1371/journal.pone.0278095
- [19] M. R. Hossain, "Predicting customer churn in telecommunications with machine learning models," *Asian J. Res. Comp. Sci.*, vol. 18, no. 1, pp. 53–66, 2025. doi: 10.9734/ajrcs/2025/v18i1400
- [20] AMTOB / AT Capital Research, "Bangladesh telecoms sector: challenges & opportunities," White Paper, 2010. [Online]. Available: <http://www.amtob.org.bd>